

Exercise 2: Implementing the Factory Method Pattern

Scenario:

You are developing a document management system that needs to create different types of documents (e.g., Word, PDF, Excel). Use the Factory Method Pattern to achieve this.

Steps:

1. Create a New Java Project:

- Create a new Java project named **FactoryMethodPatternExample**.

2. Define Document Classes:

- Create interfaces or abstract classes for different document types such as **WordDocument**, **PdfDocument**, and **ExcelDocument**.

3. Create Concrete Document Classes:

- Implement concrete classes for each document type that implements or extends the above interfaces or abstract classes.

4. Implement the Factory Method:

- Create an abstract class **DocumentFactory** with a method **createDocument()**.
- Create concrete factory classes for each document type that extends **DocumentFactory** and implements the **createDocument()** method.

5. Test the Factory Method Implementation:

- Create a test class to demonstrate the creation of different document types using the factory method.

CODE:

Document.java

```
J Document.java > Document
1 public interface Document {
2     void open();
3 }
```

ExcelDocument.java

```
J ExcelDocument.java > ExcelDocument
1 public class ExcelDocument implements Document {
2     public void open() {
3         System.out.println(x: "Opening Excel document...");
4     }
5 }
```

PdfDocument.java

```
J PdfDocument.java > PdfDocument > open()
1 public class PdfDocument implements Document {
2     public void open() {
3         System.out.println(x: "Opening PDF document...");
4     }
5 }
```

WordDocument.java

```
J WordDocument.java > WordDocument
1 public class WordDocument implements Document {
2     public void open() {
3         System.out.println(x: "Opening Word document...");
4     }
5 }
```

DocumentFactory.java

```
J DocumentFactory.java > DocumentFactory
1 public abstract class DocumentFactory {
2     public abstract Document createDocument();
3 }
```

ExcelDocumentFactory.java

```
J ExcelDocumentFactory.java > ExcelDocumentFactory
1 public class ExcelDocumentFactory extends DocumentFactory {
2     public Document createDocument() {
3         return new ExcelDocument();
4     }
5 }
```

PdfDocumentFactory.java

```
J PdfDocumentFactory.java > PdfDocumentFactory
1 public class PdfDocumentFactory extends DocumentFactory {
2     public Document createDocument() {
3         return new PdfDocument();
4     }
5 }
```

WordDocumentFactory.java

```
J WordDocumentFactory.java > WordDocumentFactory
1 public class WordDocumentFactory extends DocumentFactory {
2     public Document createDocument() {
3         return new WordDocument();
4     }
5 }
```

Main.java

```
J Main.java > Main > main(String[])
1 public class Main {
    Run | Debug
2     public static void main(String[] args) {
3         DocumentFactory wordFactory = new WordDocumentFactory();
4         Document wordDoc = wordFactory.createDocument();
5         wordDoc.open();
6
7         DocumentFactory pdfFactory = new PdfDocumentFactory();
8         Document pdfDoc = pdfFactory.createDocument();
9         pdfDoc.open();
10
11        DocumentFactory excelFactory = new ExcelDocumentFactory();
12        Document excelDoc = excelFactory.createDocument();
13        excelDoc.open();
14    }
15 }
```

OUTPUT:

```
● PS C:\Users\sunke\Desktop\Exercise 2 Implementing the Factory Method Pattern> javac Main.java
● PS C:\Users\sunke\Desktop\Exercise 2 Implementing the Factory Method Pattern> java Main
Opening Word document...
Opening PDF document...
Opening Excel document...
```